

Multi-Agent Custom Automation Engine

An open-source Azure solution accelerator that deploys coordinated AI agents to automate complex, multi-department business workflows — from a single natural-language task description.

5+	3	Azure	1-cmd
Validated Industry Use Cases	Specialized Agent Domains (MCP)	Native Cloud Deployment	azd up Deployment

PURPOSE & PROBLEM STATEMENT

Organizations face mounting pressure to automate cross-departmental workflows — onboarding new hires, reviewing contracts, coordinating marketing launches — each requiring input from multiple teams. Traditional point solutions demand a custom app per workflow, creating silos, duplicated effort, and fragile integrations.

MACAE solves this with a **single, reusable orchestration platform**. Users describe what they need in plain language; the engine recruits the right AI agents, builds a plan, executes each step, and validates the result — with full auditability.

KEY BENEFITS

- **Process Efficiency** — automates repetitive coordination tasks end-to-end
- **Error Reduction** — multi-agent validation catches mistakes before completion
- **Resource Optimization** — humans focus on judgment; agents handle execution
- **Scalability** — one platform handles unlimited workflow types without rebuilding
- **GenAI Adoption** — reduces friction of deploying AI across the organization
- **Cost Efficiency** — usage-based Azure pricing with no fixed per-workflow cost

SOLUTION ARCHITECTURE

A three-tier, fully containerized architecture deployed on Microsoft Azure:

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Frontend	React / TypeScript Fluent UI v9	Chat UI for task input & monitoring
Backend	Python / FastAPI Azure AI Agent SDK	Orchestration & agent coordination
MCP Server	Python / FastMCP	Domain tools exposed to agents
Data	Azure Cosmos DB	Tasks, plans, audit history
AI Engine	Azure OpenAI (GPT-4 class)	Agent reasoning & generation
Infra	Container Apps Key Vault · Managed ID	Hosting, secrets, identity

AGENTIC FLOW

1. **Orchestrator Agent** receives the task and generates an execution plan.
2. **Specialized Sub-Agents** execute plan steps using MCP domain tools.
3. **Validation Layer** cross-checks results before surfacing output.
4. **Full audit trail** persisted to Cosmos DB for traceability.

VALIDATED USE CASES

Use Case	Persona	Challenge	MACAE Outcome
Employee Onboarding	HR Manager	Disconnected steps: IT setup, paperwork, compliance training require manual coordination and cause delays.	Agents automate each onboarding step in sequence — ID card, payroll, benefits, mentor assignment — in minutes.
Product Marketing Launch	Marketing Executive	Release plans require sign-off from Engineering, Design, and Compliance, leading to misalignment and delays.	Dynamic agent team drafts timelines, assigns owners, and validates compliance automatically.
RFP / Contract Review	VP Finance / Legal	Manual review under tight deadlines misses compliance gaps and slows decision-making.	AI agents review documents, flag risks, recommend remediation, and log decisions with full traceability.
Retail Customer Remediation	Customer Success Mgr	Fragmented data, manual workflows, and difficulty engaging the right teams slow resolution.	Agents analyze satisfaction signals, identify root cause, and orchestrate remediation steps across teams.
Contract Compliance	Compliance Counsel	Reviewing contracts for regulatory compliance is labor-intensive and error-prone.	Agents scan contracts, surface violations, generate remediation plans, and maintain audit logs.

TECHNOLOGY STACK

Backend: Python 3.x · FastAPI · Azure AI Agent SDK (Magentic) · FastMCP

Frontend: React 18 · TypeScript · Fluent UI v9 · Vite · Axios

AI / LLM: Azure OpenAI Service (GPT-4 class) · Azure AI Foundry

Data: Azure Cosmos DB (NoSQL)

Infrastructure: Azure Container Apps · Container Registry · Key Vault · Managed Identity

IaC / Deploy: Bicep · Azure Developer CLI (azd) · GitHub Actions

Testing: pytest · vitest · Dev Containers · GitHub Codespaces

Extensibility: MCP (Model Context Protocol) — plug in any domain tool

SECURITY & COMPLIANCE

- Azure Managed Identity — no passwords or secrets in code
- Azure Key Vault — all service credentials stored and rotated centrally
- Azure AD authentication — optional per-endpoint enforcement
- GitHub secret scanning recommended for all forks
- Optional WAF + VNet integration for Container Apps
- Microsoft Responsible AI transparency documentation included

DEPLOYMENT & COST

Deployed via **azd up** (Azure Developer CLI ≥ 1.18) in a single command. Supported regions include East US, East US2, UK South, Sweden Central, Japan East. Cost is primarily **usage-based** (OpenAI tokens, Cosmos DB RUs, Container App compute) with a small fixed fee for Azure Container Registry. Resources are fully teardown-able with **azd down**.

GETTING STARTED

- ① Clone the repo ② Run **azd up** ③ Open the web UI ④ Submit a task in plain language

Supports: GitHub Codespaces, VS Code Dev Containers, local setup.

Open Source on GitHub · github.com/microsoft/Multi-Agent-Custom-Automation-Engine-Solution-Accelerator